

Microscopic Traffic Simulation: Calibration and Dynamic applications

Task 2: Dynamic Applications

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Overview

1. Dynamic dwell times
2. Public transport prioritization
3. Preliminary results
4. Overall comparison

DYNAMIC DWELL TIMES

Dynamic Dwell Times Logic

The dwell time at bus stops is modeled dynamically based on passenger demand using the formula:

$$t_{\text{dwell}(s)} = 15 + 0.5 \cdot (n_{\text{board}(s)} + n_{\text{alight}(s)})$$

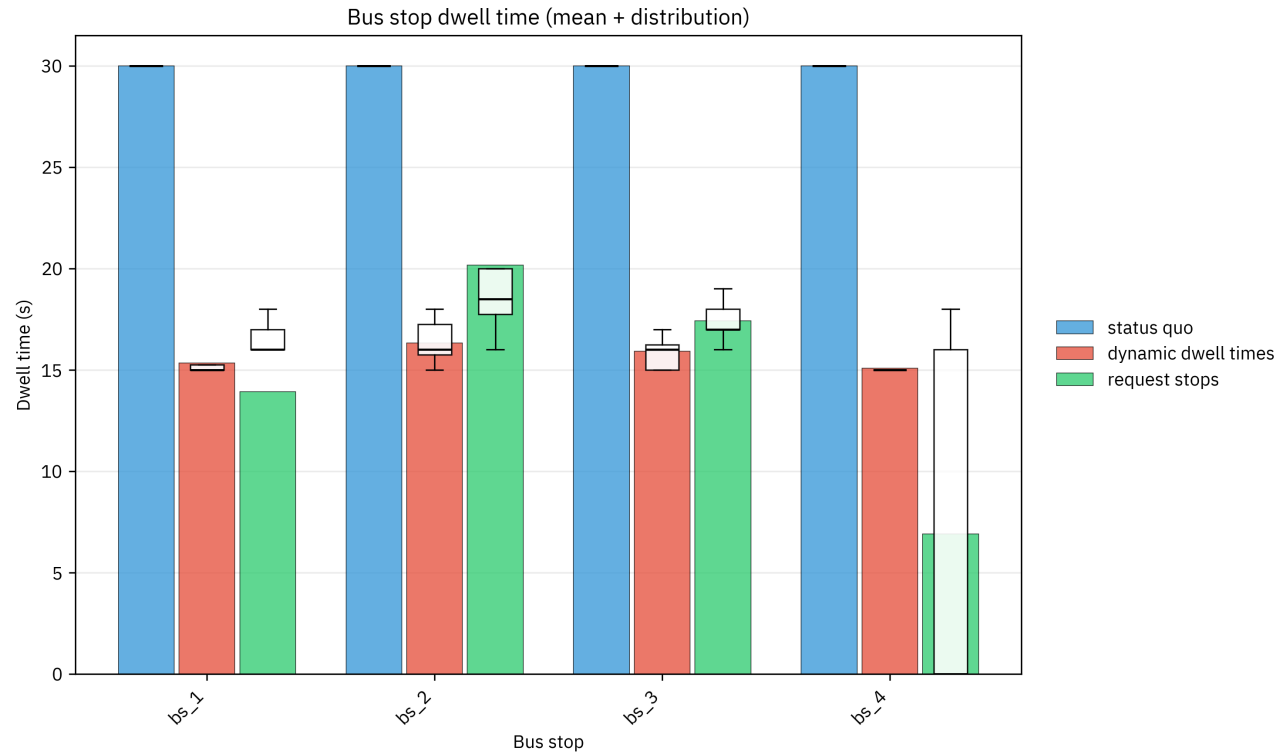
Implementation: TraCI script calculates t_{dwell} and updates the simulation via `traci.vehicle.setStopParameter` with the specific duration.

Request Stops Logic

To optimize travel times, a “Request Stop” logic was implemented **on top** of the dynamic dwell times. At a distance of 50-60m from each stop:

- **Detection:** Using `traci.vehicle.getNextStops`.
- **Decision:** The stop is skipped if no passengers are waiting (`traci.busstop.getPersonCount`) and no passengers on board intend to alight.

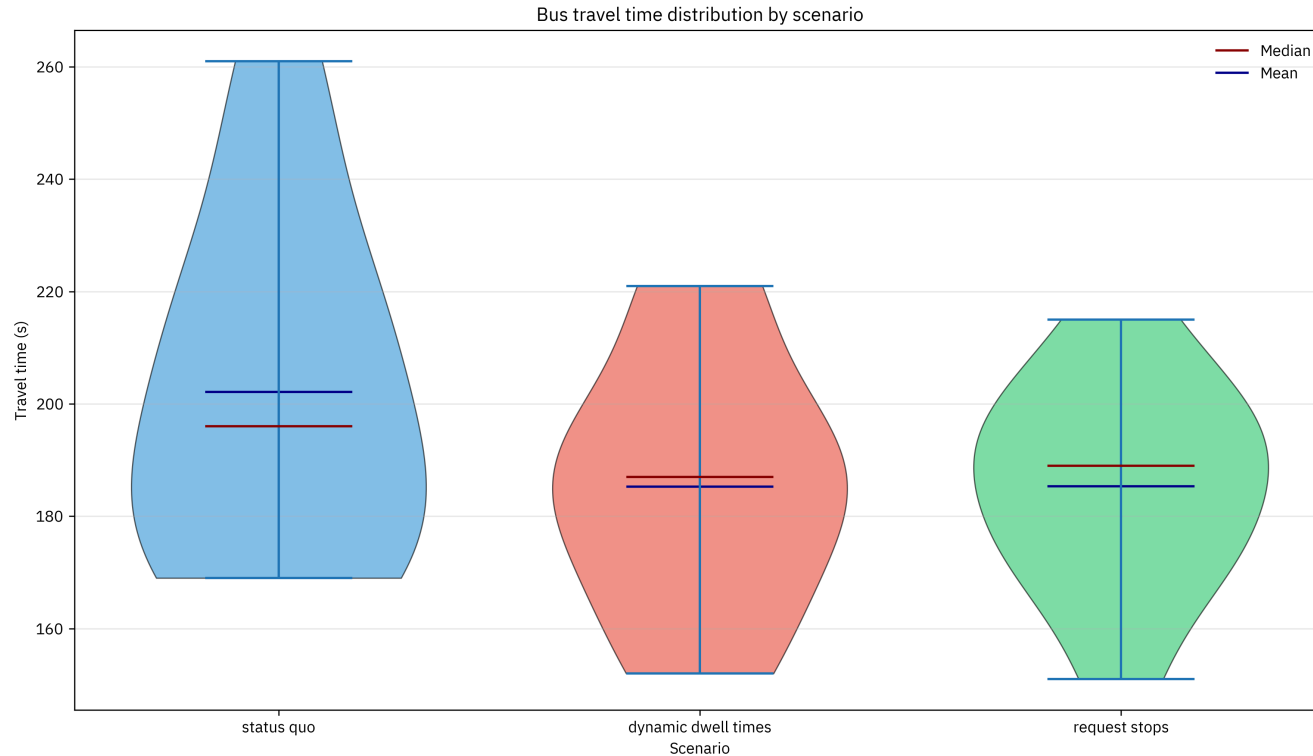
Comparison with status quo - dwell times



Dynamic dwell times:
significant dwell time
reduction vs fixed
dwell times.

Request stops: lower
average dwell times at
stops with low
passenger activity.

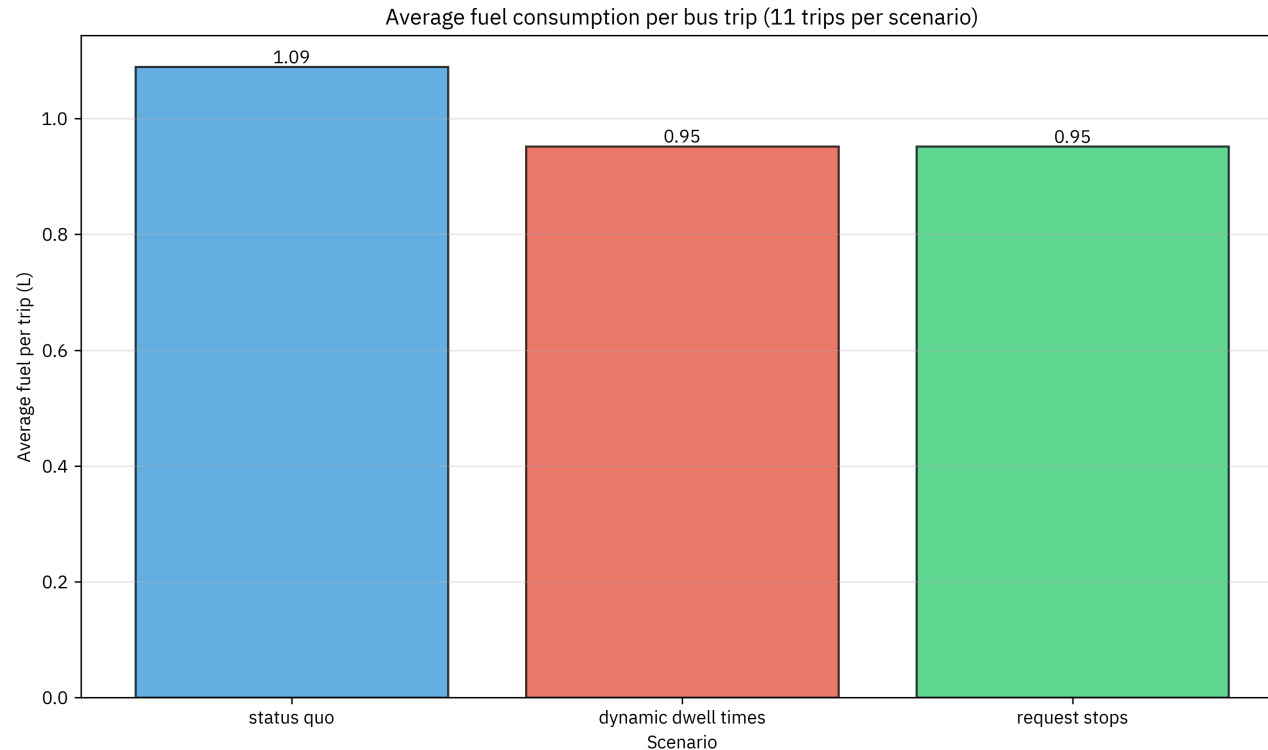
Comparison with status quo - travel times



Both dynamic dwell times and request stops lead to reduced travel times across all routes.

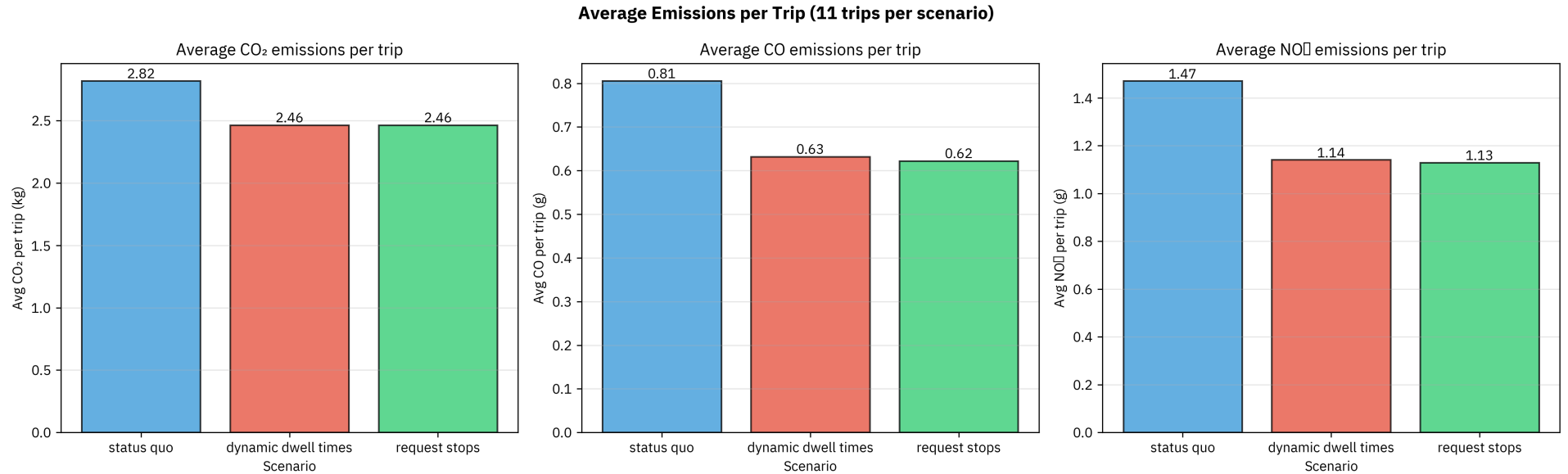
However: request stops logic does not show additional benefits.

Comparison with status quo - fuel consumption



Improved dwell times logic leads to reduced fuel consumption (lower idle time).

Comparison with status quo - emissions



Average emissions show the same behavior as the fuel consumption.

PUBLIC TRANSPORT PRIORITIZATION

PT Prioritization Logic

Real-time, rule-based control logic implemented for a sample junction (Krefelder Straße / Monheimsallee) to prioritize buses.

Problem: absolute priority

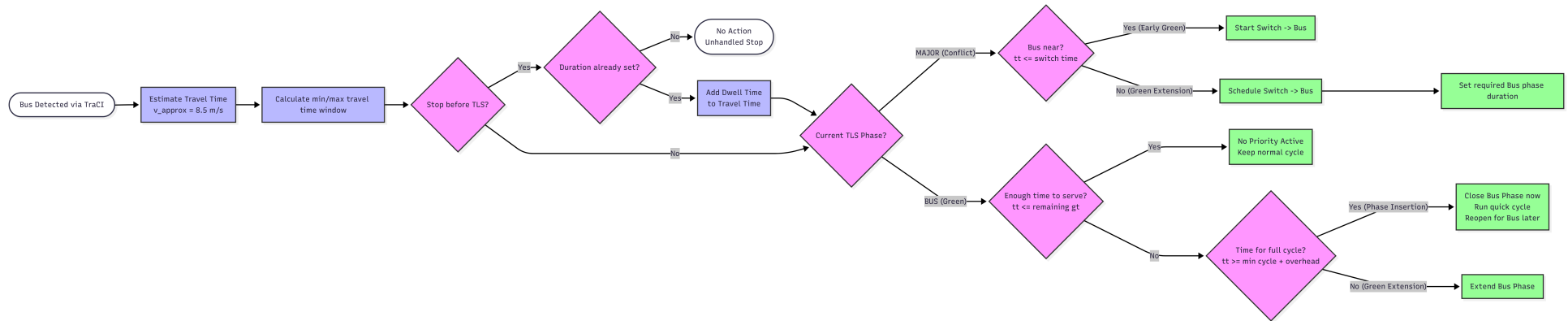
Absolute priority significantly reduces transit delay but introduces **cycle deviation** and can disrupt overall traffic flow and coordinated signal timings.

Solution: dual-stage strategy

1. **Active Intervention:** Dynamic phase adjustment (Early Green / Extension) for approaching buses.
2. **Offset Recovery:** Real-time resyncing to restore baseline signal timings post-priority.

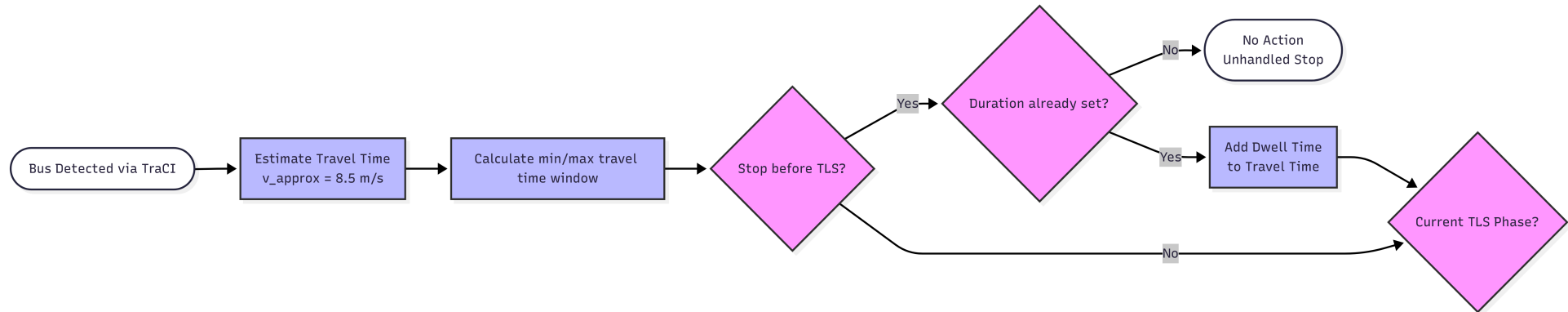
Prioritization Algorithm

Quite complex set of rules based on the bus position, stop duration, and signal state.



Prioritization Algorithm - Detection

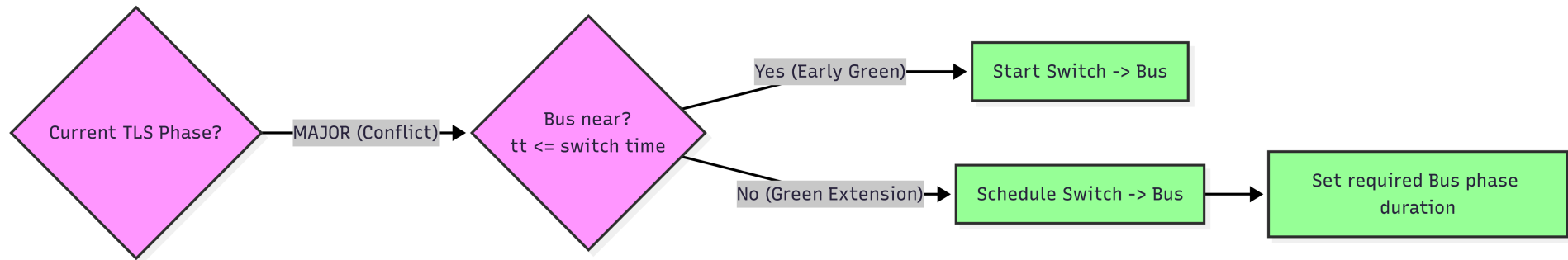
Detection step: loop until a bus is detected within the 200m approach zone **AND** its stop duration is set.



Prioritization Algorithm - Major phase

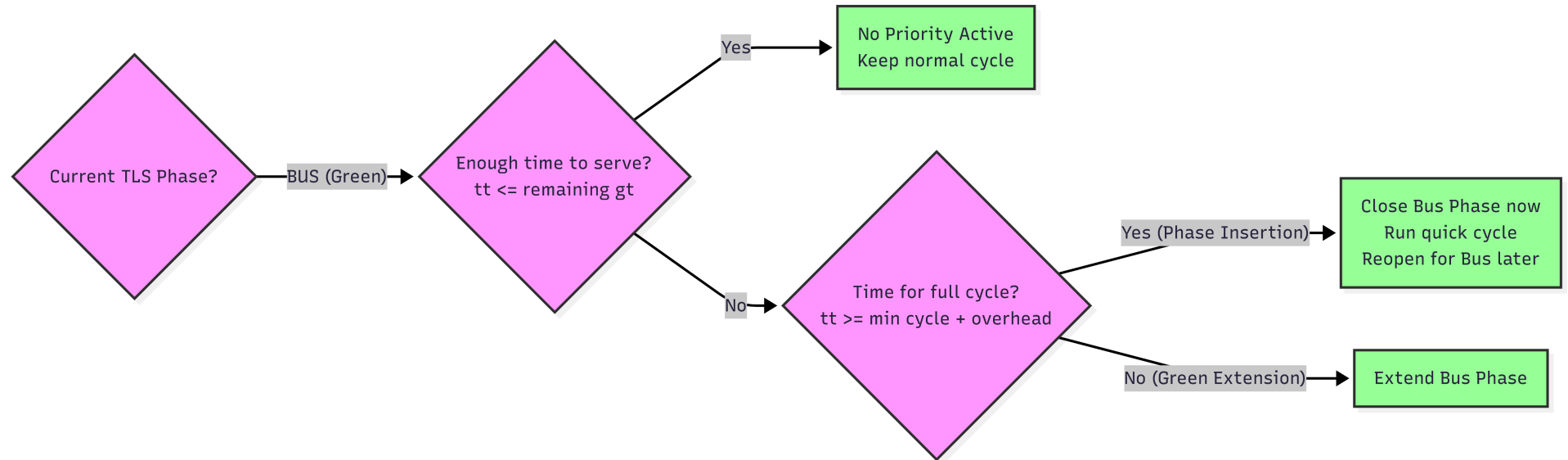
Major phase: use the time needed to switch to the bus phase as the main decision metric.

If there is enough time, schedule the switch instead of executing it immediately (reduces impact on opposing traffic stream).



Prioritization Algorithm - Bus phase

Bus phase: decide between early green, green extension, or running a quick cycle based on the remaining green time and bus arrival time.



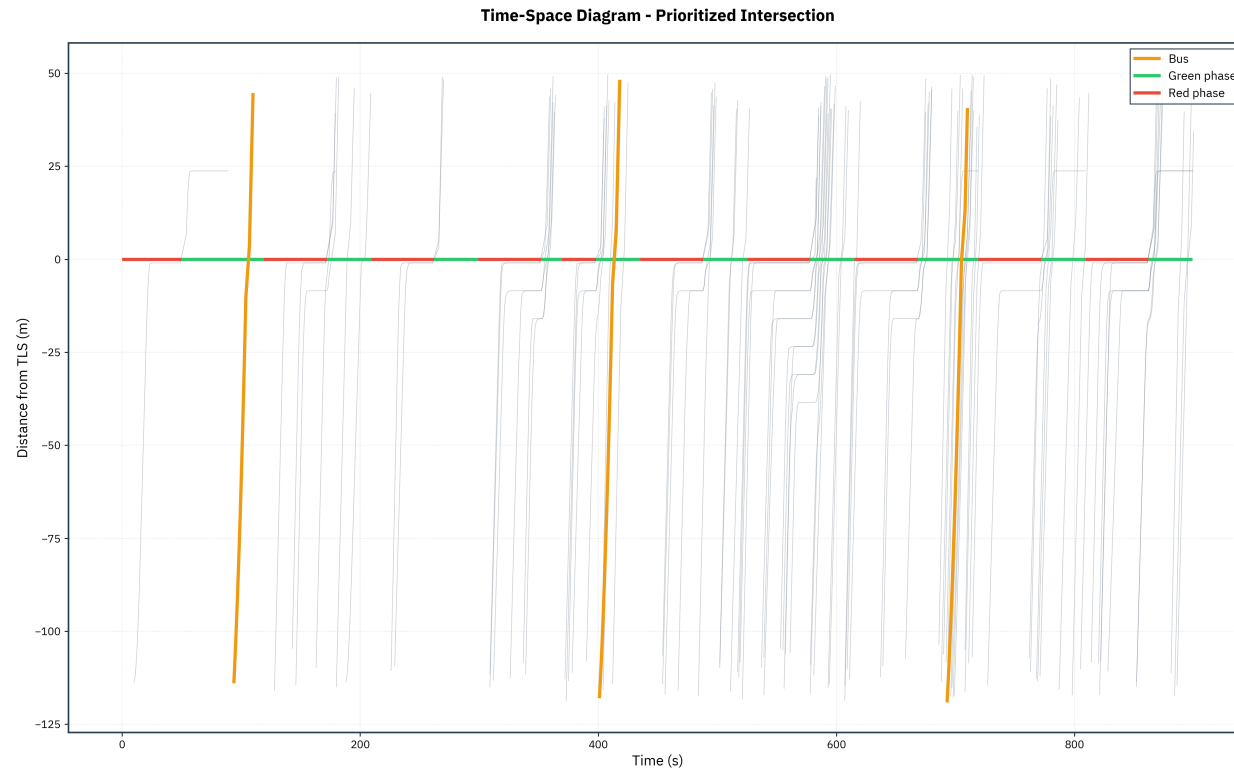
Signal Resyncing Strategy

After prioritization, a distributed resyncing strategy is applied to gradually restore the original signal timings over multiple cycles.

- **Drift Calculation:** $t_{\text{diff}} = t_{\text{actual}} - t_{\text{reference}}$.
- **Bi-phase Recovery:** The drift is compensated by compressing the “Bus” phase and extending the “Major” phase.
- **Constraints:** Durations are never reduced below the minimum green times.

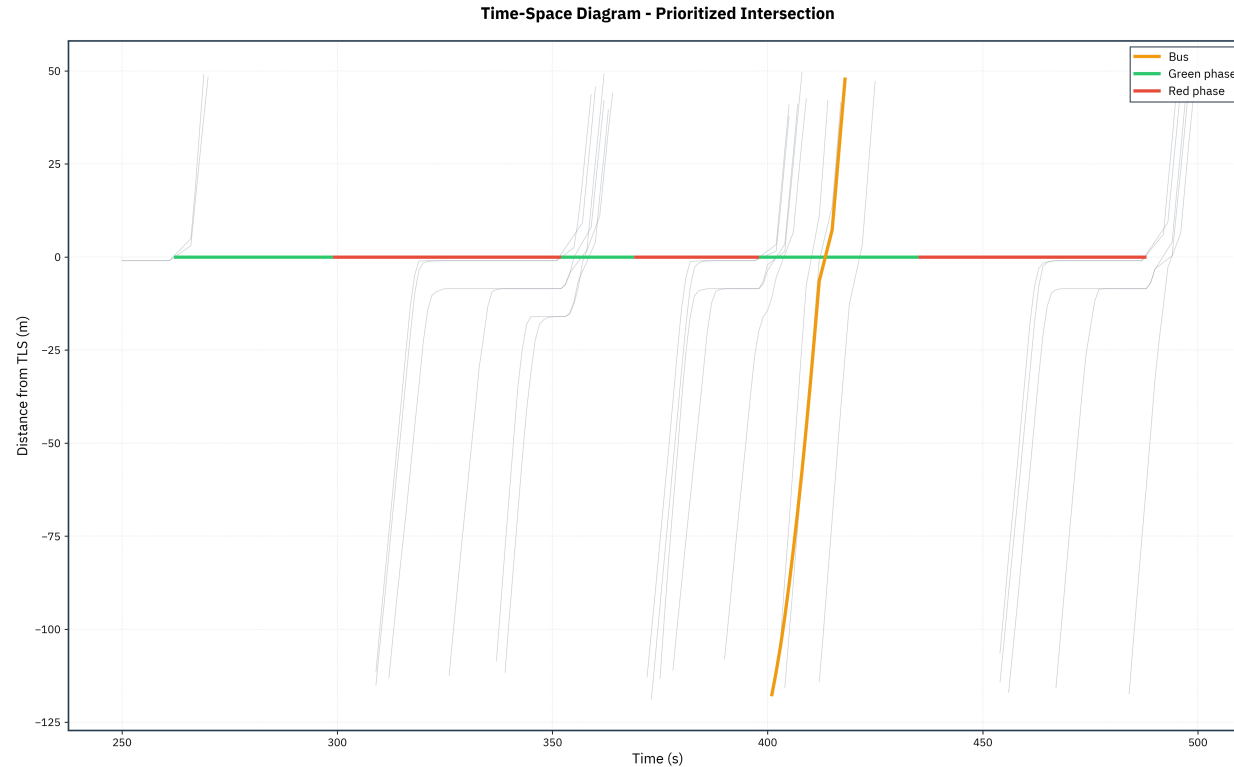
PRELIMINARY RESULTS

Time-space diagram of prioritized junction



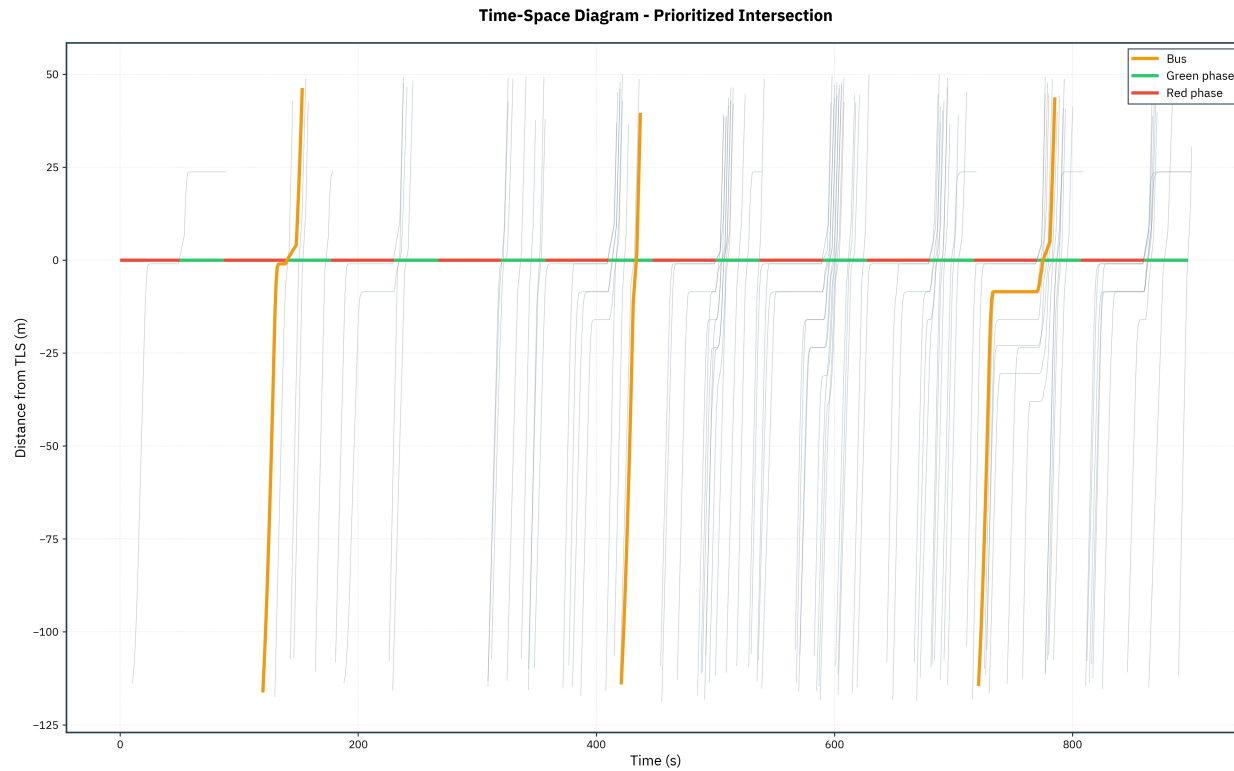
Initial testing shows effectiveness of the prioritization logic.

Time-space diagram of prioritized junction



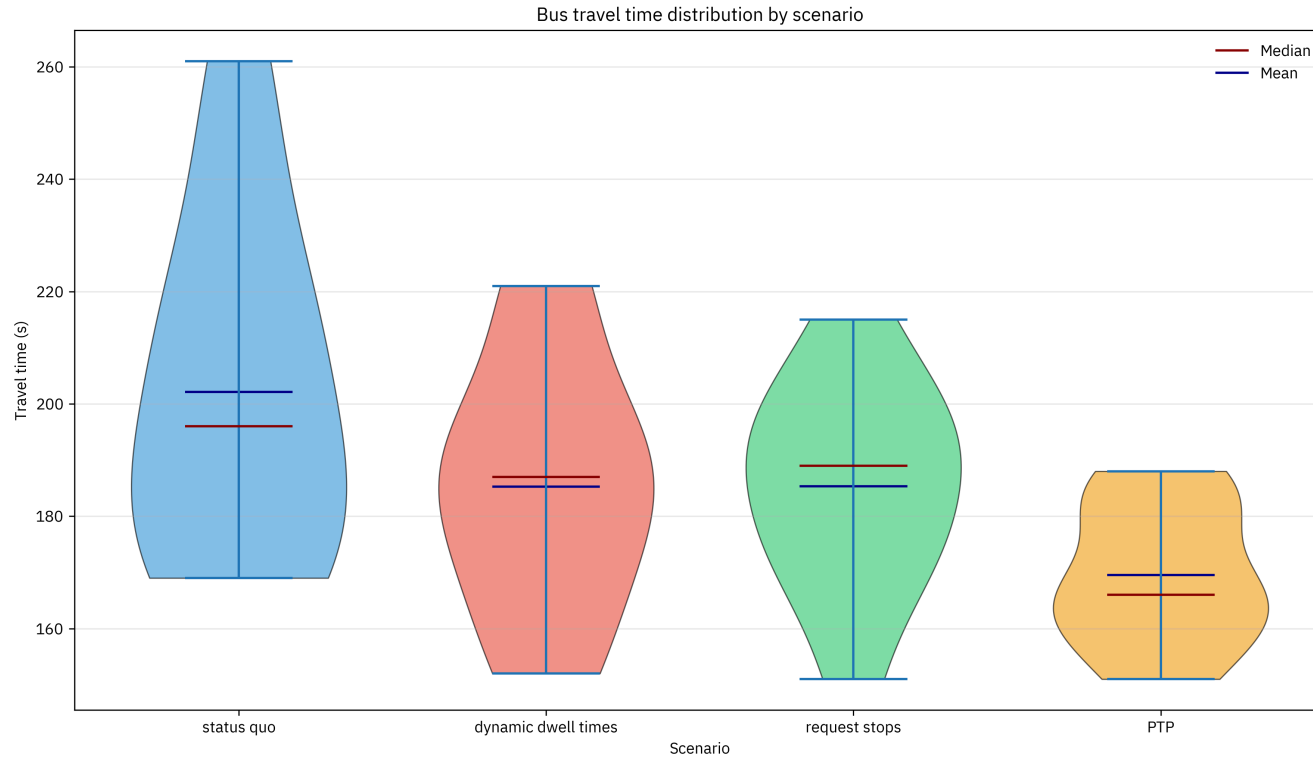
Initial testing shows effectiveness of the prioritization logic.

Time-space diagram of prioritized junction - status quo



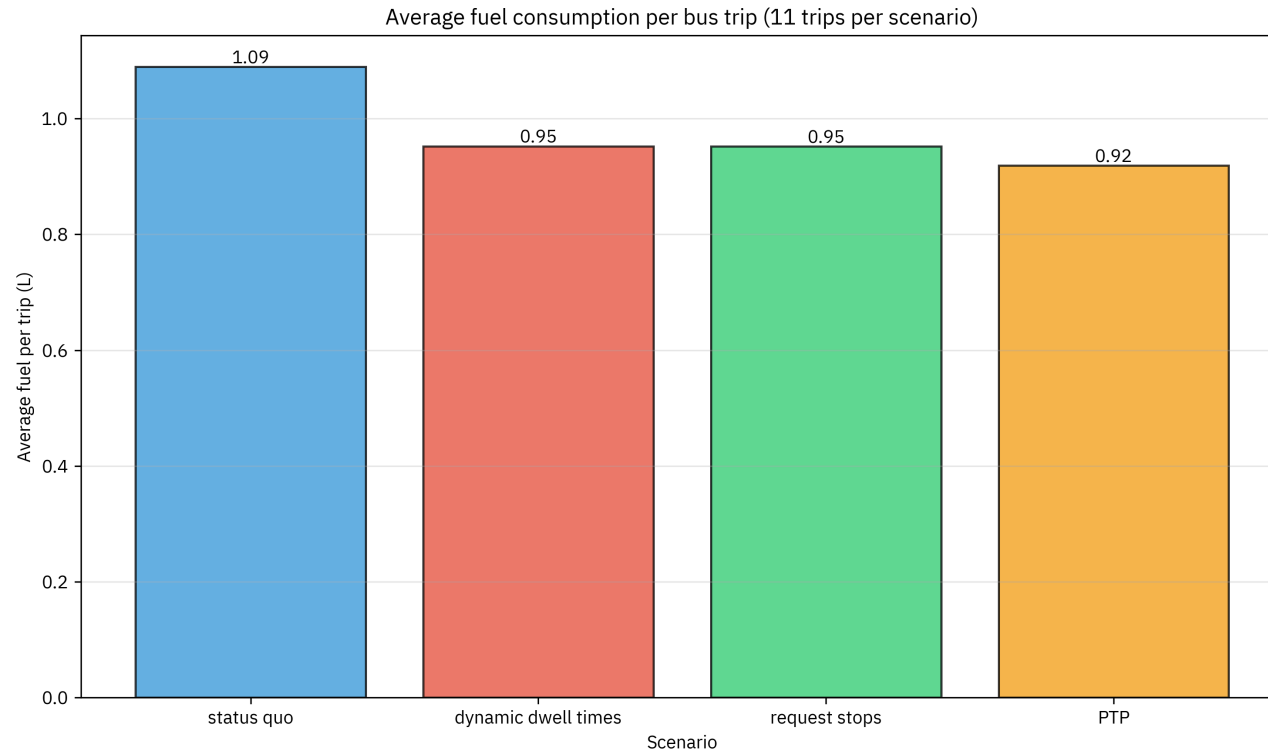
OVERALL COMPARISON

Comparison - travel times



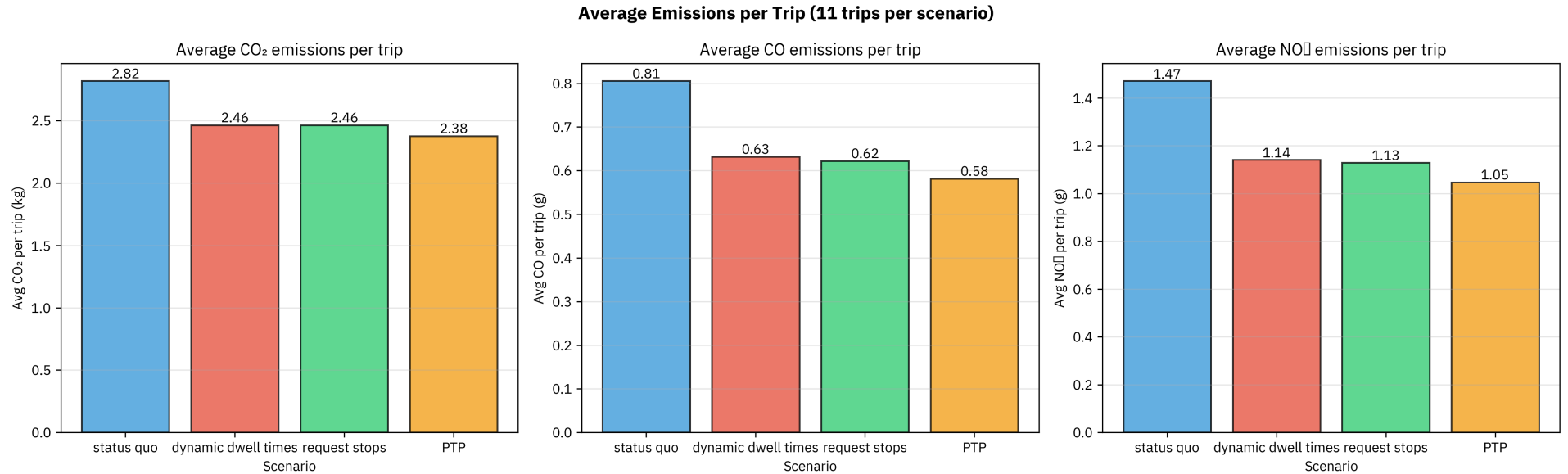
The addition of public transport prioritization further reduces travel times compared to only dynamic dwell times and request stops.

Comparison - fuel consumption



The addition of public transport prioritization further reduces fuel consumption compared to only dynamic dwell times and request stops.

Comparison - emissions



Average emissions show the same behavior as the fuel consumption.

Conclusion

The use of **dynamic dwell times** significantly improves bus performance in terms of travel times and fuel consumption.

The implementation of **request stops** did not yield additional benefits in a reduced simulation setup.

Public transport prioritization already shows significant potential in both travel time and fuel consumption reduction - even after only being implemented at a single intersection.

Thank you!